

Table: Movies

Id	Title	Director	Year	Length_minutes
1	Toy Story	John Lasseter	1995	81
2	A Bug's Life	John Lasseter	1998	95
3	Toy Story 2	John Lasseter	1999	93
4	Monsters, Inc.	Pete Docter	2001	92
5	Finding Nemo	Andrew Stanton	2003	107
6	The Incredibles	Brad Bird	2004	116
7	Cars	John Lasseter	2006	117
8	Ratatouille	Brad Bird	2007	115
9	WALL-E	Andrew Stanton	2008	104
10	Up	Pete Docter	2009	101

`SELECT * FROM movies;`

Exercise 1 — Tasks

- Find the **title** of each film
- Find the **director** of each film
- Find the **title** and **director** of each film
- Find the **title** and **year** of each film
- Find **all** the information about each film

Stuck? Read this task's [Solution](#).
Solve all tasks to continue to the next lesson.

RESET

Finish above Tasks

SELECT Title FROM Movies;

SELECT Director FROM Movies;

SELECT Title, Director FROM Movies;

SELECT Title, Year FROM Movies;

SELECT * FROM Movies;

Table: Movies

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1	Toy Story	John Lasseter	1995	81
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3	Toy Story 2	John Lasseter	1999	93
4	Monsters, Inc.	Pete Docter	2001	92
5	Finding Nemo	Andrew Stanton	2003	107
6	The Incredibles	Brad Bird	2004	116
7	Cars	John Lasseter	2006	117
8	Ratatouille	Brad Bird	2007	115
9	WALL-E	Andrew Stanton	2008	104
10	Up	Pete Docter	2009	101

Exercise 2 — Tasks

- Find the movie with a row **id** of 6
- Find the movies released in the **year** s between 2000 and 2010
- Find the movies **not** released in the **year** s between 2000 and 2010
- Find the first 5 Pixar movies and their release **year**

SELECT * FROM Movies WHERE Id = 6;

SELECT * FROM Movies WHERE Year BETWEEN 2000 AND 2010;

SELECT * FROM Movies WHERE Year NOT BETWEEN 2000 AND 2010;

SELECT Title, Year FROM Movies ORDER BY Year ASC LIMIT 5;

Id	Title	Director	Year	Length_minutes
1	Toy Story	John Lasseter	1995	81
2	A Bug's Life	John Lasseter	1998	95
3	Toy Story 2	John Lasseter	1999	93
4	Monsters, Inc.	Pete Docter	2001	92
5	Finding Nemo	Andrew Stanton	2003	107
6	The Incredibles	Brad Bird	2004	116
7	Cars	John Lasseter	2006	117
8	Ratatouille	Brad Bird	2007	115
9	WALL-E	Andrew Stanton	2008	104
10	Up	Pete Docter	2009	101

Exercise 3 — Tasks

1. Find all the Toy Story movies
2. Find all the movies directed by John Lasseter
3. Find all the movies (and director) not directed by John Lasseter
4. Find all the WALL-* movies

SELECT * FROM Movies WHERE Title LIKE 'Toy Story%';

SELECT * FROM Movies WHERE Director = 'John Lasseter';

SELECT * FROM Movies WHERE Director != 'John Lasseter';

SELECT * FROM Movies WHERE Title LIKE 'WALL-%';

Id	Title	Director	Year	Length_minutes
1	A Bug's Life	John Lasseter	1998	95
2	Toy Story	John Lasseter	1995	81
3	Monsters University	Dan Scanlon	2013	110
4	Finding Nemo	Andrew Stanton	2003	107
5	Toy Story 2	John Lasseter	1999	93
6	The Incredibles	Brad Bird	2004	116
7	Monsters, Inc.	Pete Docter	2001	92
8	Cars 2	John Lasseter	2011	120
9	WALL-E	Andrew Stanton	2008	104
10	Up	Pete Docter	2009	101

Exercise 4 — Tasks

1. List all directors of Pixar movies (alphabetically), without duplicates
2. List the last four Pixar movies released (ordered from most recent to least)
3. List the **first** five Pixar movies sorted alphabetically
4. List the **next** five Pixar movies sorted alphabetically

SELECT DISTINCT Director FROM Movies ORDER BY Director ASC;

SELECT * FROM Movies ORDER BY Year DESC LIMIT 4;

SELECT * FROM Movies ORDER BY Title ASC LIMIT 5;

SELECT * FROM Movies ORDER BY Title ASC LIMIT 5 OFFSET 5;

City	Country	Population	Latitude	Longitude
Guadalajara	Mexico	1500800	20.659699	-103.349609
Toronto	Canada	2795060	43.653226	-79.383184
Houston	United States	2195914	29.760427	-95.369803
New York	United States	8405837	40.712784	-74.005941
Philadelphia	United States	1553165	39.952584	-75.165222
Havana	Cuba	2106146	23.05407	-82.345189
Mexico City	Mexico	8555500	19.432608	-99.133208
Phoenix	United States	1513367	33.448377	-112.074037
Los Angeles	United States	3884307	34.052234	-118.243685
Ecatepec de Morelos	Mexico	1742000	19.601841	-99.050674

Review 1 — Tasks

1. List all the Canadian cities and their populations
2. Order all the cities in the United States by their latitude from north to south
3. List all the cities west of Chicago, ordered from west to east
4. List the two largest cities in Mexico (by population)
5. List the third and fourth largest cities (by population) in the United States and their population

SELECT City, Population FROM North_american_cities WHERE Country = 'Canada';

SELECT * FROM North_american_cities WHERE Country = 'United States' ORDER BY Latitude DESC;

SELECT * FROM North_american_cities WHERE Longitude < -87.6298 ORDER BY Longitude ASC;

SELECT * FROM North_american_cities WHERE Country = 'Mexico' ORDER BY Population DESC LIMIT 2;

SELECT City, Population FROM North_american_cities WHERE Country = 'United States' ORDER BY Population DESC LIMIT 2 OFFSET 2;

Id	Title	Director	Year	Length_minutes
1	Toy Story	John Lasseter	1995	81
2	A Bug's Life	John Lasseter	1998	95
3	Toy Story 2	John Lasseter	1999	93
4	Monsters, Inc.	Pete Docter	2001	92
5	Finding Nemo	Andrew Stanton	2003	107
6	The Incredibles	Brad Bird	2004	116
7	Cars	John Lasseter	2006	117
8	Ratatouille	Brad Bird	2007	115
9	WALL-E	Andrew Stanton	2008	104
10	Up	Pete Docter	2009	101

Movie_id	Rating	Domestic_sales	International_sales
5	8.2	380843261	555900000
14	7.4	268492764	475066843
8	8	206445654	417277164
12	6.4	191452396	368400000
3	7.9	245852179	239163000
6	8	261441092	370001000
9	8.5	233000154	287500000

Query Results

Id	Title	Director	Year	Length_minutes
1	Toy Story	John Lasseter	1995	81
2	A Bug's Life	John Lasseter	1998	95
3	Toy Story 2	John Lasseter	1999	93
4	Monsters, Inc.	Pete Docter	2001	92
5	Finding Nemo	Andrew Stanton	2003	107
6	The Incredibles	Brad Bird	2004	116
7	Cars	John Lasseter	2006	117
8	Ratatouille	Brad Bird	2007	115
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Exercise 6 — Tasks

1. Find the domestic and international sales for each movie
2. Show the sales numbers for each movie that did better internationally rather than domestically
3. List all the movies by their ratings in descending order

SELECT Movies.Title, Boxoffice.Domestic_sales, Boxoffice.International_sales FROM Movies JOIN Boxoffice ON Movies.Id = Boxoffice.Movie_id;

```

SELECT Movies.Title, Boxoffice.Domestic_sales, Boxoffice.International_sales
FROM Movies
JOIN Boxoffice ON Movies.Id = Boxoffice.Movie_id
WHERE Boxoffice.International_sales > Boxoffice.Domestic_sales;

```

```

SELECT Movies.Title, Boxoffice.Rating
FROM Movies
JOIN Boxoffice ON Movies.Id = Boxoffice.Movie_id
ORDER BY Boxoffice.Rating DESC;

```

Table: Buildings (Read-Only)		Table: Employees (Read-Only)			
Building_name	Capacity	Role	Name	Building	Years_employed
1e	24	Engineer	Becky A.	1e	4
1w	32	Engineer	Dan B.	1e	2
2e	16	Engineer	Sharon F.	1e	6
2w	20	Engineer	Dan M.	1e	4
		Engineer	Malcom S.	1e	1
		Artist	Tylar S.	2w	2
		Artist	Sherman D.	2w	8
		Artist	Jakob J.	2w	6
		Artist	Lillia A.	2w	7
		Artist	Brandon J.	2w	7

Query Results			
Role	Name	Building	Years_employed
Engineer	Becky A.	1e	4
Engineer	Dan B.	1e	2
Engineer	Sharon F.	1e	6
Engineer	Dan M.	1e	4
Engineer	Malcom S.	1e	1
Artist	Tylar S.	2w	2
Artist	Sherman D.	2w	8
Artist	Jakob J.	2w	6
Artist	Lillia A.	2w	7
Artist	Brandon J.	2w	7

Exercise 7 — Tasks			
1. Find the list of all buildings that have employees			
2. Find the list of all buildings and their capacity			
3. List all buildings and the distinct employee roles in each building (including empty buildings)			

```

SELECT DISTINCT Building
FROM Employees;

```

```

SELECT Building_name, Capacity
FROM Buildings;

```

```

SELECT Buildings.Building_name, Buildings.Capacity, Employees.Role
FROM Buildings
LEFT JOIN Employees ON Buildings.Building_name = Employees.Building
GROUP BY Buildings.Building_name, Employees.Role;

```

Table: Buildings (Read-Only)		Table: Employees (Read-Only)			
Building_name	Capacity	Role	Name	Building	Years_employed
1e	24	Engineer	Becky A.	1e	4
1w	32	Engineer	Dan B.	1e	2
2e	16	Engineer	Sharon F.	1e	6
2w	20	Engineer	Dan M.	1e	4
		Engineer	Malcom S.	1e	1
		Artist	Tylar S.	2w	2

Query Results			
Role	Name	Building	Years_employed
Engineer	Becky A.	1e	4
Engineer	Dan B.	1e	2
Engineer	Sharon F.	1e	6
Engineer	Dan M.	1e	4
Engineer	Malcom S.	1e	1
Artist	Tylar S.	2w	2
Artist	Sherman D.	2w	8
Artist	Jakob J.	2w	6
Artist	Lillia A.	2w	7
Artist	Brandon J.	2w	7

- Exercise 8 — Tasks
- Find the name and role of all employees who have not been assigned to a building
 - Find the names of the buildings that hold no employees

```
SELECT Name, Role
FROM Employees
WHERE Building IS NULL;
```

```
SELECT Buildings.Building_name
FROM Buildings
LEFT JOIN Employees ON Buildings.Building_name = Employees.Building
WHERE Employees.Building IS NULL;
```

Table: Movies (Read-Only)					Table: Boxoffice (Read-Only)			
Id	Title	Director	Year	Length_minutes	Movie_id	Rating	Domestic_sales	International_sales
1	Toy Story	John Lasseter	1995	81	5	8.2	380843261	555900000
2	A Bug's Life	John Lasseter	1998	95	14	7.4	268492764	475066843
3	Toy Story 2	John Lasseter	1999	93	8	8	206445654	417277164
4	Monsters, Inc.	Pete Docter	2001	92	12	6.4	191452396	368400000
5	Finding Nemo	Andrew Stanton	2003	107	3	7.9	245852179	239163000
6	The Incredibles	Brad Bird	2004	116	6	8	261441092	370001000

Query Results				
Id	Title	Director	Year	Length_minutes
1	Toy Story	John Lasseter	1995	81
2	A Bug's Life	John Lasseter	1998	95
3	Toy Story 2	John Lasseter	1999	93
4	Monsters, Inc.	Pete Docter	2001	92
5	Finding Nemo	Andrew Stanton	2003	107
6	The Incredibles	Brad Bird	2004	116
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- Exercise 9 — Tasks
- List all movies and their combined sales in **millions** of dollars
 - List all movies and their ratings in **percent**
 - List all movies that were released on even number years

```

SELECT Movies.Title,
       (Boxoffice.Domestic_sales + Boxoffice.International_sales) / 1000000 AS
Combined_Sales_Millions
FROM Movies
JOIN Boxoffice ON Movies.Id = Boxoffice.Movie_id;

```

```

SELECT Movies.Title,
       (Boxoffice.Rating * 10) AS Rating_Percent
FROM Movies
JOIN Boxoffice ON Movies.Id = Boxoffice.Movie_id;

```

```

SELECT Movies.Title, Movies.Year
FROM Movies
WHERE Movies.Year % 2 = 0;

```

Role	Name	Building	Years_employed
Engineer	Becky A.	1e	4
Engineer	Dan B.	1e	2
Engineer	Sharon F.	1e	6
Engineer	Dan M.	1e	4
Engineer	Malcom S.	1e	1
Artist	Tylar S.	2w	2
Artist	Sherman D.	2w	8
Artist	Jakob J.	2w	6
Artist	Lillia A.	2w	7
Artist	Brandon J.	2w	7

Exercise 10 — Tasks

1. Find the longest time that an employee has been at the studio
2. For each role, find the average number of years employed by employees in that role
3. Find the total number of employee years worked in each building

```

SELECT MAX(Years_employed) AS Longest_Time
FROM Employees;

```

```

SELECT Role, AVG(Years_employed) AS Average_Years
FROM Employees
GROUP BY Role;

```

```

SELECT Building, SUM(Years_employed) AS Total_Years_Worked
FROM Employees
GROUP BY Building;

```